

BricolR: Rasters with R

An Introduction

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Contents

1. The basics of rasters with R

What is a raster, how to create one, plot it and see its main features

2. Raster operations with R

Crop, operate, merge etc...

3. Raster and vectors interaction

Mask, extract, rasterize and polygonize

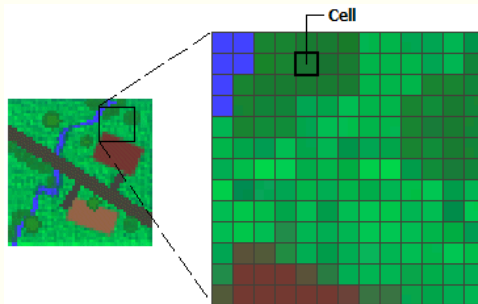
4. Can GDAL help me go faster?

Sharing my trial and errors with GDAL

The basic of rasters with R

What is a raster?

1. It's basically an image where each pixel has a **value**
2. It has an origin, an extent and a resolution



Source :<http://desktop.arcgis.com>

Rasters with R

- ❖ You can load an existing raster from a file or create a blank one with specified extent, origin and resolution...
- ❖ The rasters imported rasters are not stored in your environment
- ❖ Access their characteristics either through RStudio GU, dedicated functions (RES,ORIGIN,EXTENT) or @ORIGIN...
- ❖ access basic stats with cellStats

Plotting Rasters

- ❖ The raster package has a inbuilt `PLOT` function that works like the basic plot function with R
- ❖ You can
 - ❖ change the title with `MAIN`
 - ❖ change the color scale with `COL` and add a palette `WES.PALETTE(N=...,NAME=...)`
 - ❖ have a discrete scale with `BREAKS`
 - ❖ change the labels with `XLAB,YLAB`
 - ❖ remove the axes with `AXES=F`

Plotting Rasters with ggplot

If you prefer it is also possible to use ggplot to plot your data. Considering it requires to store the dataframe of the values, I do not recommend to use this for large datasets

1. transform your raster into a dataframe with cells coordinates : `as.data.frame(raster,xy=T)`
2. plot this dataframe with `GEOM_RASTER()` and other classic ggplot functions

Raster operations with R

Manipulate rasters

- ❖ To do operations on 2 rasters, you need to have the same grid. You can use :
 - ❖ `projectRaster`
 - ❖ `aggregate`
 - ❖ `resample`
- ❖ You can apply basic operations to rasters directly $x+y$ or produce binary rasters if you want
- ❖ You can also use the `CALC()` function that will be better for large datasets

Merge and crop

- ❖ You can merge to datasets with the function `MERGE`. Both rasters need to have the same origin and the same resolution. You can also use the `MOSAIC` function.
- ❖ If you want to look at the data on a specific location, you can use the `crop` function
- ❖ You can also mask along a polygon to only retain the data within the polygon

Raster and vectors interaction

From vectors to rasters

- ▣ You can use the function rasterize shapefiles with either RASTERIZE or ST_RASTERIZE (although I don't quite know how to do it)

stars object

I have learnt last Friday that stars object existed and should be used. Unfortunately I don't know very well how to... We should discover it together!

From raster to vectors

- ❖ You can use the functions `polygonize`, `rasterToContour`, `rasterToPoints`

Extract values from a raster depending on regions

- ❖ The extract function allows us to aggregate the values for regions given by a geometry

Storing your work

- ❖ RasterLayers are virtual, you need to save them before with `WRITERASTER`
- ❖ You can save different rasters as a raster brick or a stack and then save them as a multilayer file (can be useful for analysis or for plotting it in QGIS)
- ❖ You can also store your working environment with `SAVE.IMAGE()`

Using GDAL

Making the best of GDAL

I have only started using GDAL but I think, with large datasets, it can really make the things go much faster. Let's see that together.

Thank You!